

Akif Ejaz

akifejaz40@gmail.com | linkedin.com/in/akifejaz | akifejaz.dev

EXPERIENCE

Sr. Engineer

Mar. 2025 – Present

10xEngineers

Lahore, Pakistan

- Leading strategic collaboration with the Eclipse Foundation to architect and scale RISC-V support for the ThreadX RTOS, driving architectural clean-ups, feature enhancements, and long-term maintainability for both RV64 and RV32 ports.
- Enabled SeL4 Microkernel support for the SpacemiT's BananaPi BPI-F3 RISC-V board by implementing the low-level kernel initialization, peripheral drivers (UART, Timer etc) and validating functionality with the full SeL4 Tests suite achieving 100% pass rate.
- Leading the maintenance and horizontal scaling of Cloud-V (RISC-V CI Services Platform), increasing system uptime to 97.6% and supporting 120+ global users across diverse CI pipelines, ensuring seamless developer operations and driving platform reliability.
- Evaluated the performance of multiple RISC-V SBCs from SiFive, SpacemiT, and StarFive using SPEC, CoreMark, and Phoronix suites, publishing 10+ technical comparison reports on Cloud-V platform to provide the RISC-V community with data-driven hardware insights and simplify board selection.

Software Engineer - II

Mar. 2023 – Feb 2025

Xcelerium Inc.

California, USA

- Developed custom board support (BSP) package for Xcelerium's RISC-V64 core in Zephyr RTOS and enabled Xcelerium's proprietary RISC-V instructions including RISC-V Vector Extension.
- Led the "Linux Boot" project, successfully achieved the goal of booting Linux Kernel v6.10 on the Xcelerium's RISC-V SW simulator (Hydra VP) and Scalar Core (Hydra Tile SU).
- Developed a Sequence Pytest framework using pytest framework for the verification of RISC-V Vector processor having RVV 1.0 and Xcelerium's proprietary instructions support.
- Contributed in SW RISC-V Simulator by adding the RVV Cryptography instructions to support RVV Crypto Extension v1.0.0.
- Engineered a high-performance AWS verification suite by integration with FPGA and DUT memory maps, refactoring data transfer protocols from 8-bit to 64-bit widths to achieve a 700% increase in load/store throughput.

EDUCATION

Information Technology University

Lahore, Pakistan

Bachelor of Science in Computer Engineering (BSCE)

CGPA 3.76/4.00

ACHIEVEMENTS, CONTRIBUTIONS & MORE

Eclipse ThreadX Committer | I'm official Eclipse ThreadX RTOS committer (see [here](#)). Working on improving RISC-V support and making it RISC-V first RTOS. See my contributions on [ThreadX GitHub](#) Feb 2026

ATESOR Developer & Maintainer | ATESOR is open-source multi-stage agentic framework for autonomous RISC-V software porting. It is based on LangGraph self driven smart agentic system with Docker and Native RISC-V based test environments. See more about it on [ATESOR GitHub](#) March 2026

SeL4 Microkernel Maintainer | I added BananaPi BPI-F3 board support in seL4 microkernel and since then I'm officially maintaining its support (see maintained by status [here](#)). Also see my contributions on [seL4 GitHub](#) Dec 2025

Author in Spectrum of Emerging Sciences | I had privilege to get my paper accepted in Spectrum of Emerging Sciences journal, titled "An Efficient ML & DL Based Deep Packet Security Framework For Detection of Computing Network Faults In The IOTs". Read paper on [Zenodo](#) May 2025

TECHNICAL SKILLS

Languages: C, C++, Python, Bash/Shell Scripting, System Verilog, Verilog, RISC-V Assembly, Linker scripts, Makefiles, Git (CI/CD)

Embedded OS/Kernel Dev.: Linux Kernel, Microkernel, KVM, Zephyr, FreeRTOS, TinyML, TensorFlow

Frameworks: Pytest, UVM, Cocotb, Selenium

Developer Tools: Git, CI/CD, Vivado, Verdi, Spike, Google Colab, VS Code, Matlab

Libraries: Pandas, NumPy, Matplotlib